TOPIC 1 · THE GLOBAL ECONOMY

Source: The Conversation (Anita Talberg, Univ. of Melbourne), 11 Nov 2026
Wednesday 11 November 2026

Why COP31 wants 35% of the world's energy to be electric by 2035

COP31's '**35-by-35**' pledge, led by Chris Bowen and Türkiye's Murat Kurum, asks countries to get **35% of global final energy** from electricity by 2035, up from **23%** today. **Anita Talberg** explains the number.

WHAT HAPPENED

- **Why 35%?** It is the **median** level of electrification in the ten lowest-warming climate scenarios: 23% (2025), 28% (2030) and 35% (2035).
- **Off track:** the IEA projects that current policies deliver just under **30%** by 2035.
- **1.5°C will be overshot:** the best case is a peak of about **1.7°C** in the 2040s, falling back to 1.5°C by 2100.
- **Efficiency:** electric cars and heat pumps use energy far more efficiently than fossil-fuel equivalents. Up to **three-quarters** of energy demand could be electrified with existing technology.
- **Barriers:** grid and storage investment, incentives to switch from gas, and affordable finance.
- **Energy security:** countries that import oil and gas were exposed to this year's price spike from the US–Iran conflict.

Final energy: energy reaching users such as households and factories
Overshoot: warming temporarily exceeding 1.5°C before being brought back down
Subsidy: a government payment that lowers producers' costs

KEY NUMBERS

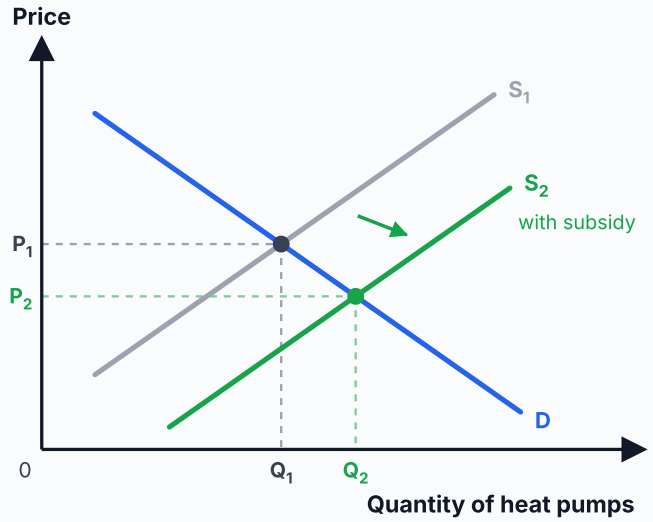
23% → 35%
Electricity share of global final energy, 2025 to 2035 target

~30%
IEA projection for 2035 on current policies

1 in 4
New cars sold globally in 2025 that were electric

~1.7°C
Best-case peak warming, 2040s

DIAGRAM — Speeding the switch: subsidising electric technology



Governments can speed electrification by **subsidising** electric equipment such as heat pumps, which lowers producers' costs. Supply shifts from **S_1 to S_2**.

The price falls from **P_1 to P_2** and take-up rises from **Q_1 to Q_2**, displacing gas appliances.

The subsidy has a **budget cost** and an **opportunity cost**, so it is justified only if the climate and energy-security benefits are larger.

Exam angle: cheaper finance and subsidies help **developing economies** electrify. This links global environmental goals with development, and is one reason climate finance is central at COPs.

BAR CHART — Electricity's share of global final energy



Source: Climate Resource (2026) analysis of climate scenarios; IEA. Bars scaled to 40%.

SYLLABUS LINK

Topic 1 – global environmental issues and international cooperation (COP31). Topic 4 – environmental management policy (subsidies, regulation). Topic 2 – Australia's energy security and reliance on imported fuel.

TRY THIS

Explain how government policies could speed up electrification, and assess one economic cost of doing so.
(4 MARKS)

The Conversation, Anita Talberg – The world needs to get serious about electrification – here's why (11 Nov 2026). Uploaded by the teacher. Disclosure: the author's organisation, Climate Resource, received Australian Government funding for analysis on electrification as a COP31 priority.

Diagrams are illustrative, not to scale